

Math 540
Homework 5

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Section 51

Exercise 1. Show that if $h, h' : X \rightarrow Y$ are homotopic and $k, k' : Y \rightarrow Z$ are homotopic, then $k \circ h$ and $k' \circ h'$ are homotopic.

Proof. Since $h \cong h'$, there exists a continuous function $F : X \times I \rightarrow Y$ such that $F(x, 0) = h(x)$ and $F(x, 1) = h'(x)$. Since $k \cong k'$, there exists a continuous function $G : Y \times I \rightarrow Z$ such that $G(y, 0) = k(y)$ and $G(y, 1) = k'(y)$. Then $H : X \times I \rightarrow Z$ given by $H(x, t) := G(F(x, t), t)$ is continuous as it is the composition of continuous functions. Moreover, $H(x, 0) = G(F(x, 0), 0) = G(h(x), 0) = k(h(x))$ and $H(x, 1) = G(F(x, 1), 1) = G(h'(x), 1) = k'(h'(x))$. Thus $h \circ k \cong h' \circ k'$. $\square$

Exercise 2. Given spaces X and Y , let $[X, Y]$ denote the set of homotopy classes of maps of X into Y .

- (a) Let $I = [0, 1]$. Show that for any X , the set $[X, I]$ has a single element.
- (b) Show that if Y is path connected, the set $[I, Y]$ has a single element.

Proof. (a) Let $f : X \rightarrow I$ be continuous. Let $g : X \rightarrow I$ be the constant map. Define $F : X \times I \rightarrow I$ by $F(x, t) = (1 - t)f(t) + g(x)t$. Then F is continuous because it is a linear combination of continuous functions. Moreover, $F(x, 0) = f(x)$ and $F(x, 1) = g(x)$. Therefore $f \cong g$. Since f was arbitrary, $[X, I] = \{[g]\}$.

(b) Let $f : I \rightarrow Y$ be continuous. Define $c : I \rightarrow Y$ by $c(x) = f(0)$. Since this map is constant, it is continuous. Define $F : I \times I \rightarrow Y$ by $F(x, t) = f((1 - t)x)$. Then $F(x, 0) = f(x)$ and $F(x, 1) = f(0)$. Thus $f \cong c$. Let $y \in Y$ be arbitrary. Since Y is path connected, there is an induced homotopy from the constant map $c : I \rightarrow Y$ to the constant map $g : I \rightarrow Y$ given by $g(x) = y$. Since $f \cong c$ and $c \cong g$, we have $f \cong g$. Thus $[I, Y] = \{[g]\}$. $\square$

Exercise 3. A space X is said to be contractible if the identity map $i_X : X \rightarrow X$ is nullhomotopic.

- (a) Show that I and $\mathbb{R}$ are contractible.
- (b) Show that a contractible space is path connected.
- (c) Show that if Y is contractible, then for any X , the set $[X, Y]$ has a single element.
- (d) Show that if X is contractible and Y is path connected, then $[X, Y]$ has a single element.

Proof. (a) Note that $i_{[0,1]}$ and any constant map $c : [0, 1] \rightarrow [0, 1]$ is continuous. Simply applying the straight-line homotopy gives $i_{[0,1]} \cong c$. An identical reasoning follows from $\mathbb{R}$

being contractible.

(b) Let X be contractible. There exists a continuous function $F : X \times I \rightarrow X$ such that $f(x, 0) = i_X(x) = x$ and $f(x, 1) = x_0$. Let $a, b \in X$. Define $\varphi : I \rightarrow X$ by:

$$\varphi(t) = \begin{cases} f(2t, a), & 0 \leq t \leq \frac{1}{2} \\ f(2 - 2t, b) & \frac{1}{2} \leq t \leq 1. \end{cases}$$

By the pasting lemma φ is continuous, and $\varphi(0) = a$ and $\varphi(1) = b$. Thus X is path connected.

(c) Let $f : X \rightarrow Y$ be continuous. Since Y is contractible, there exists a continuous function $G : Y \times I \rightarrow Y$ such that $G(y, 0) = i_Y(y) = y$ and $G(y, 1) = y_0$. Define $F : X \times I \rightarrow Y \times I$ by $F(x, t) = (f(x), t)$. This map is continuous, hence the composition $G \circ F : X \times I \rightarrow Y$ is continuous. Moreover, $(G \circ F)(x, 0) = f(x)$ and $(G \circ F)(x, 1) = y_0$. Thus f is nullhomotopic. Since f was arbitrary, $[X, Y]$ contains only a single element.

(d) Let $f : X \rightarrow Y$ be continuous. Since X is contractible, there exists a continuous function $F : X \times I \rightarrow X$ such that $F(x, 0) = i_X(x) = x$ and $F(x, 1) = x_0$. Since Y is path connected, there exists a continuous function $\varphi : I \rightarrow Y$ such that $\varphi(0) = f(x_0)$ and $\varphi(1) = y_0$. Note that $(f \circ F) : X \times I \rightarrow Y$ is continuous, and $(f \circ F)(x, 0) = f(x)$ and $(f \circ F)(x, 1) = f(x_0)$. Now for any $y \in Y$, there exists a path from $f(x_0)$ to y . Since f is homotopic to the constant function equal to $f(x_0)$ and since the constant function equal to $f(x_0)$ is homotopic to the constant function equal to y_0 , by transitivity f is homotopic to the constant function equal to y_0 . Thus $[X, Y]$ contains only a single element. $\square$

Section 51

Exercise 1. A subset A of $\mathbb{R}^n$ is said to be star convex if for some point a_0 of A , all the line segments joining a_0 to other points of A lie in A .

- (a) Find a star convex set that is not convex.
- (b) Show that if A is star convex, A is simply connected.

Proof. (a) A typical "star" (or its higher-dimensional analogues) is star convex. Indeed, for any point in the "star" a line segment can be drawn to its center. However, the "star" is clearly not convex.

(b) Let A be a star convex. Then there exists a point $x_0 \in A$ such that for all $a \in A$, the line segment from x_0 to a is contained in A . Let $f : [0, 1] \rightarrow A$ be a loop based at x_0 . Define $F : [0, 1] \times [0, 1] \rightarrow A$ by $F(s, t) = tx_0 + (1 - t)f(s)$. This is a homotopy between f and e_{x_0} . Thus $\pi_1(A, x_0) = \{[e_{x_0}]\}$, meaning A is simply connected. $\square$

Exercise 2. Let α be a path in X from x_0 to x_1 ; let β be a path in X from x_1 to x_2 . Show that if $\gamma = \alpha * \beta$, then $\hat{\gamma} = \hat{\beta} \circ \hat{\alpha}$.

Proof. For any $[p] \in \pi_1(X, x_0)$:

$$\begin{aligned}\hat{\beta}(\hat{\alpha}([p])) &= \hat{\beta}([\bar{\alpha} * p * \alpha]) \\ &= [\bar{b} * (\bar{\alpha} * p * \alpha * \beta)] \\ &= [(\bar{\alpha} * \beta) * p * (\alpha * \beta)] \\ &= \hat{\gamma}([p]).\end{aligned}$$

□

Exercise 3. Let x_0 and x_1 be points of the path-connect space X . Show that $\pi_1(X, x_0)$ is abelian if and only if for every pair α and β of paths from x_0 to x_1 , we have $\hat{\alpha} = \hat{\beta}$.

Proof. Suppose $\pi_1(X, x_0)$ is abelian. Then $\pi_1(X, x_1)$ is abelian as well. Observe that:

$$\begin{aligned}\hat{\alpha}([f]) &= [\bar{\alpha} * f * \alpha] \\ &= [\bar{\alpha} * f * (\beta * \bar{\beta}) * \alpha] \\ &= [(\bar{\alpha} * f * \beta) * (\bar{\beta} * \alpha)] \\ &= [(\bar{\beta} * \alpha) * \bar{\alpha} * f * \beta] \\ &= [\bar{\beta} * f * \beta] \\ &= \hat{\beta}([f])\end{aligned}$$

Thus $\hat{\alpha} = \hat{\beta}$.

Conversely, let f and g be two loops with base point x_0 and let α be a path from x_0 to x_1 . Then $\beta = f * \alpha$ is a path from x_0 to x_1 as well. By assumption, $\hat{\alpha} = \hat{\beta}$. Hence:

$$\begin{aligned}[\bar{\alpha} * g * \alpha] = [\bar{\beta} * f * \beta] &\iff [\beta \bar{\alpha} * g * \alpha] = [\bar{\beta} * f] \\ &\iff [f * g * \alpha] = [g * \beta] \\ &\iff [f * g] = [g * \beta * \bar{\alpha}] \\ &\iff [f * g] = [g * f].\end{aligned}$$

Thus $\pi_1(X, x_0)$ is abelian.

□

Exercise 5. Let A be a subspace of $\mathbb{R}^n$; let $h : (A, a_0) \rightarrow (Y, y_0)$. Show that if h is extendable to a continuous map of $\mathbb{R}^n$ into Y , then h_* is the trivial homomorphism.

Proof. Let $i : A \rightarrow \mathbb{R}^n$ be the inclusion. Extend h to a continuous function $k : \mathbb{R}^n \rightarrow Y$. Then $h = k \circ i$, hence $h_* = k_* \circ i_*$ because the fundamental group is functorial. But k_* is the trivial homomorphism because its domain is $\pi_1(\mathbb{R}^n, a_0) = \{[e_{a_0}]\}$, which is the trivial group. Thus $h_* = k_* \circ i_*$ is trivial.

□